

THE MIDNIGHT MISSION

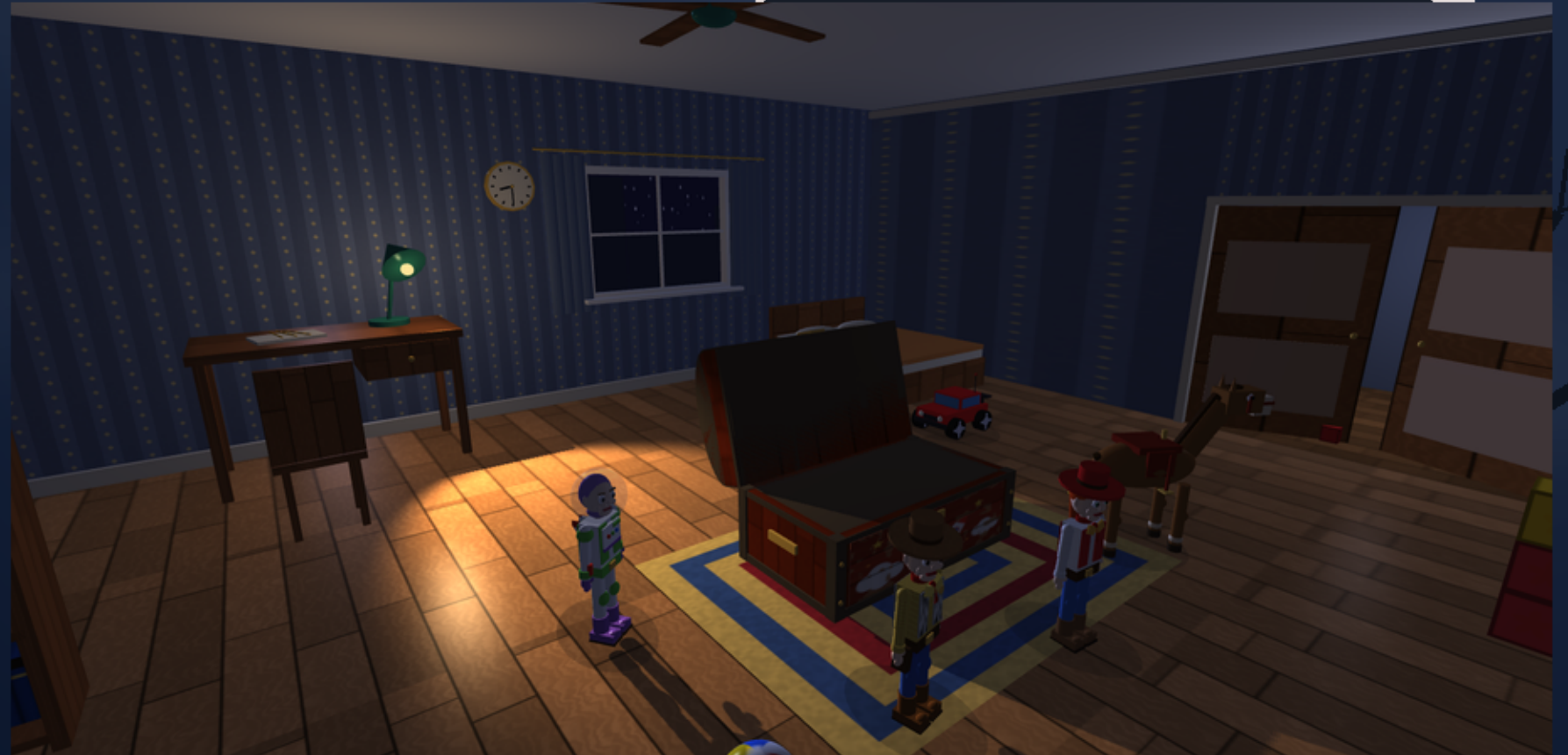
A haunted toy room. An interactive graphics story.

Adiba Tahsin · Roll 2107031

CSE-4102 · Computer Graphics and Image
Processing Laboratory

Md Tajmilur Rahman, Lecturer

Md Mubtashim Abrar Nihal, Lecturer



ROOM NIGHT

EXPLORE, CONTROL, COMPARE

The proposal is realised as an eight-stage story from night to morning.

Play as Penny or any toy; the story layer drives the rest

TRS + hierarchy; chest lid, mounting, Bullseye jump

Three light types + a sky dome outdoors

27 surface maps; analytic BVH ray tracing

PLAY PROJECT DEMO



HOUSE



TOYS ALIVE

PRIMITIVES BECOME CHARACTERS

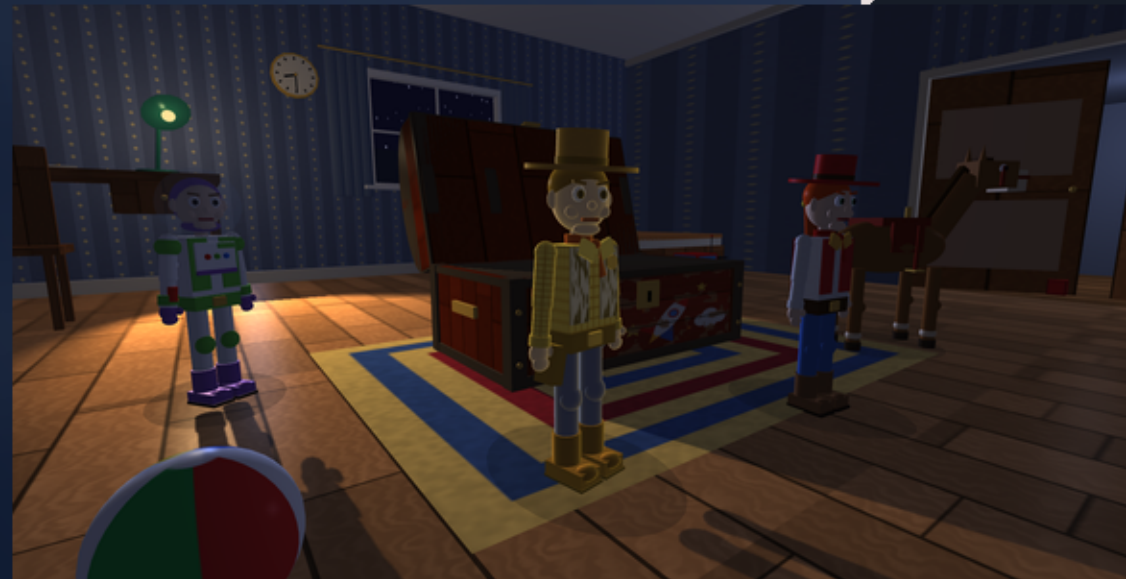
Position, normal and UV data define five indexed mesh families.

Plane 4 vertices / 2 triangles; cube 24 / 12

Sphere: $24 \times 36 \rightarrow 925$ vertices / 1656 triangles

Cylinder: 134 / 128; cone: 99 / 64

The scene uses shared buffers and instance transforms



WOODY



BUZZ



PENNY

A WORLD WITH SURFACE DETAIL

The house, garden, room and furniture belong to one connected scene.

Wood grain, wallpaper, cloth, moon craters and book rows

UV repeats multiply material colour by sampled RGB

Picket alpha cuts holes and replaces repeated geometry

27 maps plus white in one texture array serves the ray tracer



HOUSE



WINDOW



BOOKCASE

ONE PARENT MOVES THE RIDER

$$M_{\text{world}}(\text{child}) = M_{\text{world}}(\text{parent}) \times M_{\text{local}}(\text{child})$$

R: mount/dismount within 2.2 units

1–5 + W/S/A/D: choose and drive a toy

9 / PENNY button: take or release Penny

F: inspect; C: Free / Orbit / Follow

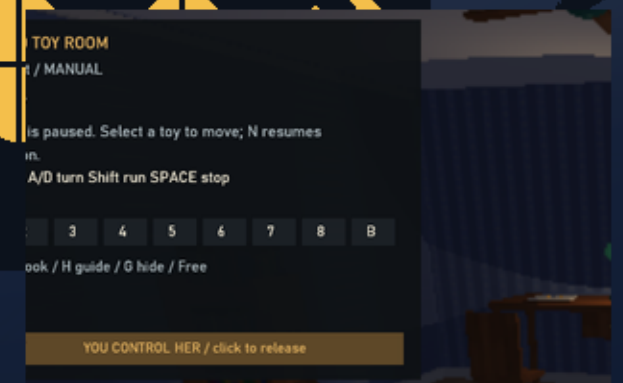
Normal matrix = inverse transpose of world linear part



MOUNTED



LIVE CONTROL



HUD PENNY PANEL

EVERY MOTION HAS A RULE

Delta time connects displacement, gait, wheel rotation and camera motion.

Gait angle = $\sin(\text{distance phase}) \times \text{amplitude}$

Hinges: chest lid, doors, wardrobe; jump
 $= 0.75 \sin(\pi t)$

Solid crowd: priority contact + steering

Spines: chains of boxes, nose to tail

Camera eases over stairs: $1 - e^{(-k\Delta t)}$



RESCUE JUMP



BUZZ FLIGHT



CROWD AFTER

THREE LIGHT TYPES

Ambient + diffuse + specular + emission

Directional: sun/moon of the sky dome

Point: lamp bulb; $1/(k_c + k_l d + k_q d^2)$

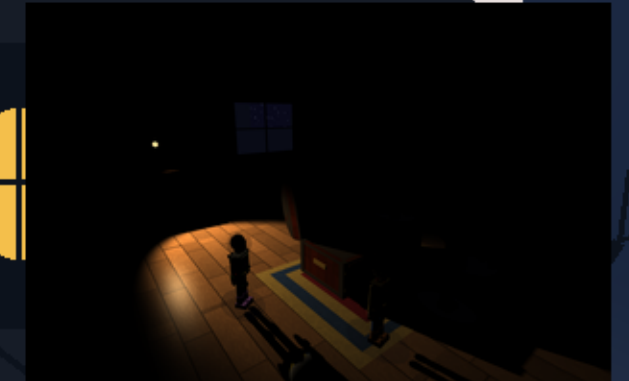
Spot: lamp/headlights; smooth inner/outer cone

Diffuse = $k_d \max(N \cdot L, 0)$

Specular = $k_s \max(R \cdot V, 0)^n$



SKY DAY RASTER



SPOT



LAMP

WHERE IS LIGHT EVALUATED?

Matched views expose the difference between vertex and pixel illumination.

Flat: triangle normal from position derivatives

Gouraud: vertex lighting → interpolated colour

Phong: interpolated normal → pixel lighting

Blinn: half-vector $H = \text{normalize}(L+V)$

F2 cycles raster modes; F4 returns to ray tracing



FLAT



GOURAUD



PHONG



BLINN

RAYS REVEAL A SECOND VIEW

Primary ray → nearest analytic hit → illumination → bounded continuation

BVH median split; world-space boxes;
near child first

Shadow rays to sun, moon and lamp

A missed ray returns the shared sky dome

Mirror floor/ball; straight transparency

Raster: lamp + sun/moon shadow maps,
PCF



ROOM RAY



SKY NIGHT RT

10 THE TOYS ARE SAFE

THANK YOU

All five escape into the morning. The world stays open.

Adiba Tahsin · Roll 2107031

CSE-4102

Geometry · hierarchy · motion ·
interaction

Lighting · shading · textures · ray tracing



STORY END

